STATEMENT OF PURPOSE

Applicant: Anthony Oluwaseun Matthew | **Academic Background:** B.Tech in Mathematics (Operations Research & Computational Mathematics), OAUSTECH

Target Program:	Master of Science (M.Sc.) in Computer Science
Target Institutions:	Eötvös Loránd University (ELTE), Budapest (1st Choice) | University of Debrecen (2nd Choice)
Official Contacts:	ceo@empirecodes.com.ng | matthew@empirecodes.com.ng | +234 817 305 7810
Engineering Portfolio:	empirecodes.com.ng | GitHub: github.com/Empirezoom

1. INTRODUCTION & ACADEMIC FOUNDATION

My journey into advanced computing is founded upon a core conviction: the most reliable, fault-tolerant software systems are fundamentally rooted in rigorous mathematical principles. Having graduated with a Bachelor of Technology in Mathematics from Olusegun Agagu University of Science and Technology (OAUSTECH), specializing in Operations Research and Computational Mathematics, I approach software engineering not merely as the syntactic assembly of code, but as the deliberate formulation of deterministic finite-state automata, discrete optimization models, and graph-theoretical algorithms.

Over the past five years, I have actively unified this rigorous mathematical background with large-scale production software engineering—architecting resilient cross-platform mobile systems, implementing zero-loss fintech payment transaction gateways, and managing decoupled cloud infrastructures. Under the prestigious Stipendium Hungaricum Scholarship Programme, I am applying for the Master of Science in Computer Science at Eötvös Loránd University (ELTE) as my first choice and the University of Debrecen as my second choice. My objective is to deepen my academic foundations in advanced distributed consensus, formal software verification, and high-performance concurrent architectures within Hungary's world-renowned informatics ecosystem.

2. BRIDGING COMPUTATIONAL MATHEMATICS & PRODUCTION ENGINEERING

Throughout my undergraduate curriculum, advanced coursework in Numerical Analysis, Linear and Non-Linear Optimization, and Discrete Mathematics demonstrated that computational complexity directly governs real-world performance. Rather than designing software heuristically, I utilized mathematical frameworks to diagnose and eliminate bottlenecks across production applications.

A concrete example arose when I engineered the real-time transaction engine for an e-commerce platform processing thousands of live transactions via the Paystack payment gateway. Under high concurrency and variable cellular latencies, standard asynchronous architectures risk race conditions and duplicate ledger postings. Applying transaction isolation theory, I engineered atomic row-level locks using select-for-update semantics at the database engine level and integrated cryptographic HMAC-SHA512 webhook signature verification. To guarantee zero data discrepancy, I designed an idempotent event-driven queueing pipeline with deterministic exponential backoff handlers. This mathematically sound design eliminated 100% of transaction state anomalies and maintained strict financial ledger consistency across unstable network conditions.

3. PROFESSIONAL EXPERIENCE & SYSTEMS ARCHITECTURE

My industry progression has reinforced my capacity to design and maintain systems across the entire decoupled stack. Through my software engineering firm, EmpireCodes, I developed and published a comprehensive production mobile and web fleet. Utilizing React Native and Expo SDK, I engineered production mobile applications incorporating offline-first state reconciliation with asynchronous storage and synchronized audio-haptic micro-interactions. On the backend, I designed high-throughput RESTful APIs using Python and Django REST Framework, containerized with Docker and deployed across cloud microservices.

Beyond building commercial products, I served as a Backend Developer and Technical Instructor at AnchorSoft Academy in Lagos, where I trained aspiring software engineers in relational database normalization, API security, and Python development. Furthermore, during my National Youth Service Corps (NYSC) deployment within the Procurement Unit of the Somolu Local Government Council (completed in 2026), I identified severe manual record-keeping bottlenecks and engineered an automated municipal budget tracking pipeline using Python and relational database schemas, reducing procurement processing turnaround by 40%.

4. WHY HUNGARY, ELTE, & CAREER OBJECTIVES

Hungary has a celebrated global reputation in mathematics, algorithmics, and computer science—from the foundational contributions of John von Neumann, Paul Erd■s, and László Lovász to today's cutting-edge informatics faculties. The Master of Science in Computer Science curriculum at Eötvös Loránd University (ELTE), Budapest, and the University of Debrecen represents the ideal intellectual bridge for my background.

I am particularly drawn to ELTE's Faculty of Informatics and its specialized tracks in Software Technology and Distributed Systems, specifically research coursework covering Formal Methods, Advanced Algorithms, and Cloud-Native Distributed Computing. Having completed my undergraduate degree in an English-speaking higher education system with proven English proficiency, I am fully equipped to thrive within Hungary's international academic community.

Following completion of my master's degree, my long-term objective is to operate as a Lead Systems Architect and Distributed Computing Researcher, contributing to mission-critical software infrastructure, advancing fintech security protocols, and fostering international technological partnerships between Europe and emerging technology ecosystems.

5. CONCLUSION

I offer the Stipendium Hungaricum selection committee and the graduate faculty at Eötvös Loránd University and the University of Debrecen a proven union of rigorous undergraduate mathematical training, five years of verified full-stack production engineering, and an unyielding commitment to academic excellence. I am prepared to dedicate myself fully to the academic rigors of the programme and to represent the values of the Stipendium Hungaricum scholarship with distinction.

Thank you for considering my application.

Respectfully submitted,

Anthony Oluwaseun Matthew
Founder & Lead Engineer, EmpireCodes
B.Tech in Mathematics (Honours), OAUSTECH
Email: ceo@empirecodes.com.ng | matthew@empirecodes.com.ng
Phone: +234 817 305 7810 / +234 813 536 2561

Portfolio: https://empirecodes.com.ng